Slide 5: Architectural Comparison

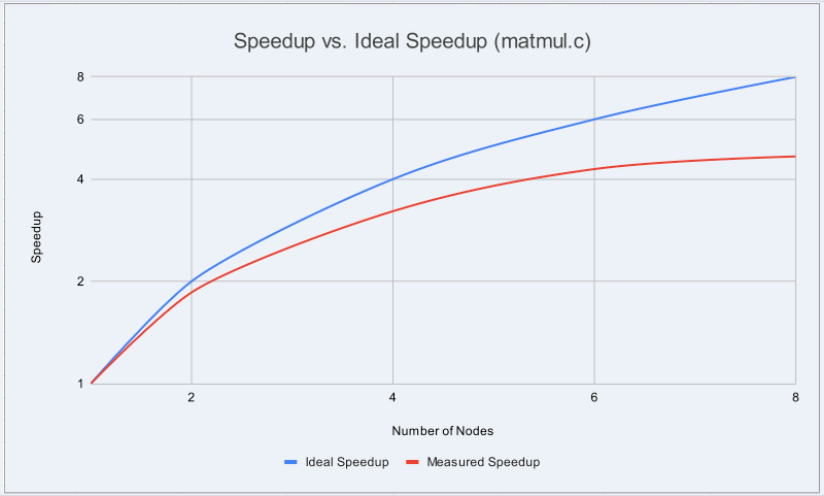
Speaker : Fuzail

Approach A MPI_ALLGATHERV

The Hypothesis: The CPU cost of generating 64 million random numbers via complex bitwise shifts is too heavy.

The Technical Implementation: Divide the PRNG generation workload proportionally across all ranks ($O(N^2/P)$). Once each node computes its slice, rely entirely on the high-speed cluster network (`MPI_Allgatherv`) to stitch the full 512 MB Matrix B together across all nodes.

Approach A Speedup Curve



Approach B ZERO-COMM

The Hypothesis: Transmitting 512 MB across a shared, multi-tenant university cluster will cause massive network congestion and unpredictable latency.

The Technical Implementation: Completely bypass the network. Accept a redundant computational penalty where every single CPU independently generates the entirety of Matrix B locally ($O(N^2)$). Achieve 0% network bandwidth consumption until the final verification gather.

Approach B Speedup Curve

